

YOUSUF ELBEYALY

Machine Learning & Generative AI Engineer

Email: yousef.amir080@gmail.comPhone: +20 112 008 4582Location: Al-Mahallah Al-Koubra, Egypt

GitHub: github.com/Yousef-ElbeyalyAvailability: Remote / Hybrid RolesLanguages: Arabic , English

PROFESSIONAL SUMMARY

Results-oriented **AI/ML & Deep Learning Engineer** with a rigorous foundation in mathematical modeling, computational statistics, and software engineering. Proven track record in developing high-performance **Computer Vision pipelines** (PyTorch, OpenCV) and architecting end-to-end **Generative AI & NLP applications** using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG) frameworks, and Vector Databases. Expert in bridging complex theoretical AI architectures—including Quantization (FP16/BFLOAT16) and Transformer decoding—into scalable, production-ready systems and interactive web deployments.

TECHNICAL SKILLS & COMPETENCIES

Generative AI & LLMs	Hugging Face Transformers , Large Language Models (T5, FLAN-T5, LLaMA), RAG Architectures , Sentence-Transformers, Prompt Engineering, Model Quantization (FP16, BFLOAT16), Fine-tuning, NLP Pipelines.
Machine & Deep Learning	PyTorch , TensorFlow, Scikit-learn, XGBoost, CatBoost, Optuna, Computational Statistics, Regression & Time-Series Forecasting, Transfer Learning (VGG16, ResNet).
Computer Vision & NLP	OpenCV, MediaPipe, NLTK, pdfplumber, Document Processing & Text Extraction, Hand Landmark Detection, Image Classification & Segmentation.
Databases & Vector Search	ChromaDB , FAISS (CPU), SQL, Vector Embeddings, Semantic Similarity Search, Cosine Similarity Indexing.
Deployment & MLOps	Streamlit , FastAPI , Uvicorn, Docker, Git, NVIDIA TensorRT, Google Cloud Platform (GCP), Hugging Face Spaces, Linux/Windows Virtual Environments.
Programming Languages	Python (Advanced, OOP, Modular Packaging), SQL, C#.

PROFESSIONAL EXPERIENCE

Freelance Machine Learning & AI EngineerJan 2026 – Present

Self-Employed | Remote Client Projects

- Architected and deployed end-to-end Machine Learning and Generative AI solutions, integrating custom Deep Learning pipelines and LLM inference engines via production-ready RESTful APIs (FastAPI, Uvicorn).
- Engineered Retrieval-Augmented Generation (RAG) document assistants, utilizing vector databases (FAISS) and sentence embeddings to enable high-accuracy semantic search and automated text summarization.
- Researched and implemented advanced theoretical deep learning architectures, including Attention mechanisms, Flow Matching, and Diffusion Models, optimizing local model performance and inference speed.

- Consulted on data leakage prevention, feature engineering, and hyperparameter tuning (Optuna) for tabular revenue forecasting and regression projects.

FEATURED ENGINEERING PROJECTS

PDF AI Assistant (End-to-End RAG & LLM System)

2026

Tech Stack: Hugging Face Transformers (T5/FLAN-T5), FAISS, Sentence-Transformers, pdfplumber, Streamlit

- Designed and deployed a local Retrieval-Augmented Generation (RAG) assistant capable of ingesting multi-page PDF reports, extracting structured text, and performing context-aware Q&A and automated chapter summarization.
- Built a high-performance semantic search pipeline using sentence-transformers and FAISS CPU index to retrieve top-k relevant document chunks in milliseconds with zero data leakage.
- Optimized local LLM decoding and inference, applying memory-efficient techniques and constructing an interactive, professional Web UI using Streamlit for seamless user ingestion and real-time chat.

Glow Up – Retail Sales & Revenue Forecasting System

2026

Tech Stack: Python, XGBoost, CatBoost, Optuna, Pandas, Time-Series Feature Engineering

- Engineered an end-to-end time-series regression system to predict daily store sales and revenue for a retail business across 289+ historical trading days, optimizing inventory planning and commercial growth.
- Executed rigorous data cleaning, temporal feature engineering, and categorical encoding while strictly eliminating data leakage (removing post-sale variables like target transaction quantity).
- Tuned gradient boosting architectures (XGBoost/CatBoost) achieving exceptional predictive accuracy and built modular evaluation pipelines to track daily revenue performance metrics.

Chest Cancer Detection & X-Ray Classification System

2025

Tech Stack: Transfer Learning (VGG16), OpenCV, Deep Learning

- Developed a high-precision Computer Vision diagnostic tool for classifying chest X-ray scans using PyTorch and Transfer Learning with deep convolutional networks (VGG16 architecture).
- Implemented advanced image preprocessing, normalization, and data augmentation techniques to prevent model overfitting and significantly enhance classification recall and F1-score on medical datasets.

AI Virtual Mouse & Real-Time Fitness Exercise Trainer

2025

Tech Stack: Python, OpenCV, MediaPipe, Real-Time HCI Processing

- Engineered a low-latency Human-Computer Interaction (HCI) system utilizing MediaPipe hand and body landmark detection to enable gesture-based mouse control without physical hardware.
- Built an automated fitness tracking module that calculates joint angles in real-time from webcam feeds, accurately counting repetitions and evaluating posture exercise mechanics.

AI Mizan – Intelligent Food Price Forecasting Engine

2025

Tech Stack: CatBoost, Scikit-learn, Flask RESTful API, Python

- Developed a predictive pricing model for food commodities utilizing CatBoost gradient boosting, achieving an outstanding 90.8% R^2 score on test datasets.
- Wrapped the trained model into a production-ready RESTful Flask API, enabling real-time pricing queries and integration with frontend consumer web applications.

CERTIFICATIONS & EDUCATION

Generative AI with Large Language Models – DeepLearning.AI / AWS

Completed

Natural Language Processing Specialization – DeepLearning.AI

Completed

Deep Learning Specialization – <i>DeepLearning.AI</i>	Completed
Machine Learning Specialization – <i>Stanford University / DeepLearning.AI</i>	Completed
Advanced Data Analytics Professional Certificate – <i>Google</i>	Completed
Bachelor's Degree in Computer Science (Faculty of Computers & Information) – <i>Mansoura University</i>	2025 – 2029 (Expected)